# Chapter 1. The Pigeon-Hole Principle

*Seven Is More Than Six*

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## 1.1 The basic pigeon-hole principle

**Motivation.** Comparing the number of objects with the number of places gives existence results; the art is choosing the balls and the boxes.

**Theorem 1.1 (Pigeon-hole principle)**

Let $n > k$ be positive integers. If $n$ identical balls are placed into $k$ identical boxes, some box receives at least two balls.

**Proof.** Otherwise every box has 0 or 1 ball, so there are at most $k < n$ balls. ■

**Example 1.2**

Some element of $7, 77, 777, \ldots$ is divisible by 2023.
*Solution.* If none of $a_1, \ldots, a_{2023}$ is divisible by 2023, their remainders lie in $\{1, \ldots, 2022\}$, so (Theorem 1.1) $2023 \mid a_j - a_i$ for some $i < j$. But $a_j - a_i = a_{j-i} \cdot 10^i$ and $\gcd(10^i, 2023) = 1$, so $2023 \mid a_{j-i}$, a contradiction.

**Example 1.3**

In a round-robin chess tournament with $n$ players, at any moment two players have finished the same number of games.
*Solution.* The possible values are $0, \ldots, n-1$, but 0 and $n-1$ cannot both occur (a player with $n-1$ games has played everybody). So at most $n-1$ values occur; apply Theorem 1.1.

## 1.2 The generalized pigeon-hole principle

**Motivation.** Many more balls than boxes force some box to contain many balls.

**Theorem 1.4 (Pigeon-hole principle, general version)**

Let $n, m, r$ be positive integers with $n > rm$. If $n$ identical balls are distributed into $m$ identical boxes, some box receives at least $r + 1$ balls.

**Proof.** Otherwise there are at most $rm < n$ balls. ■

**Example 1.5**

Ten points lie in a unit square. Then two of them are at distance $< 0.48$, and three of them can be covered by a disk of radius 0.5.
*Solution.* Nine squares of side $\frac{1}{3}$: one contains two points (Theorem 1.1), at distance $\leq \frac{\sqrt{2}}{3} < 0.48$. Four triangles cut by the diagonals: since $10 > 2 \cdot 4$, one contains three points (Theorem 1.4), and its circumradius is $\frac{1}{2}$.

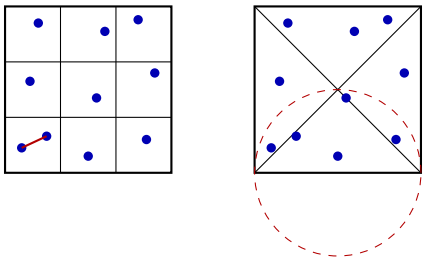


**Figure.** *Example 1.5: nine squares give two close points; four triangles give three points under a disk of radius* $\frac{1}{2}$.

**Example 1.6**

Within the last 1000 years, the reader had an ancestor $A$ such that some person $P$ was an ancestor of both parents of $A$.
*Solution.* 40 generations of 25 years give a family tree with $2^{41}-1$ nodes. If two nodes are the same person $B$, then $B$ serves as $P$ (for the nearest common descendant). Otherwise $2^{41}-1$ distinct people lived, but at most $40 \cdot 10^{10} < 2^{41}-1$ did.

**Example 1.7 (The handshake party)**

Mr. and Mrs. Smith invite four couples. Acquaintances shake hands (nobody with themself or their spouse). Afterwards Mr. Smith observes that, apart from himself, nobody shook hands the same number of times. How many hands did Mrs. Smith shake?
*Solution.* Vertices = people, edges = handshakes (a **graph**). The nine people other than Mr. Smith have distinct handshake numbers in $\{0,\dots,8\}$; let $Y_i$ have $i$.
$Y_8$ missed only the spouse, and $Y_0$ shook no hands, so $Y_8$–$Y_0$ are married.
$Y_7$ missed two people, $Y_0$ and $Y_1$ (whose only handshake was with $Y_8$), so $Y_7$–$Y_1$ are married.
Likewise $Y_6$–$Y_2$, $Y_5$–$Y_3$, so Mrs. Smith is $Y_4$: **four** handshakes.

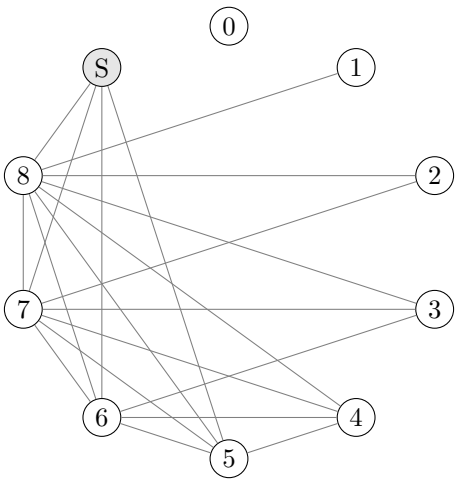



**Figure.** *The only possible party of Example 1.7 (vertex $i$ shook $i$ hands, $S$ = Mr. Smith); couples* $\{8,0\},\{7,1\},\{6,2\},\{5,3\},\{4,S\}$.

## 1.3 Selected exercises

**Motivation.** Book exercises on choosing the boxes.

**Exercise 1.11 (Sum or difference divisible by 1000)**

Prove that among any 502 positive integers there are two whose sum or difference is divisible by 1000.

**Solution.**

*Step 1 (boxes).* With $\bar{r}$ the residue class mod 1000, the 501 boxes

$$\{\bar{0}\}, \quad \{\bar{500}\}, \quad \{\bar{1}, \overline{999}\}, \quad \{\bar{2}, \overline{998}\}, \quad \ldots, \quad \{\overline{499}, \overline{501}\}$$

partition the residue classes.

*Step 2 (pigeon-hole).* $502 > 501$, so two list members $a, b$ share a box (Theorem 1.1).

*Step 3 (conclusion).* If $a \equiv b$, then $1000 \mid a - b$; otherwise $a \equiv r$, $b \equiv 1000 - r$, so $1000 \mid a + b$. ■

**Exercise 1.15 (Six goals in consecutive games)**

A soccer team played 30 games and scored 53 goals, scoring at least one goal in each game.
(a) Prove that there was a sequence of consecutive games in which the team scored exactly six goals.
(b) Prove that the claim of part (a) can fail if the team scored 60 goals, the other conditions unchanged.

**Solution.**

(a) Let $s_k$ be the goals in games $1, \ldots, k$:

$$1 \le s_1 < s_2 < \cdots < s_{30} = 53.$$

The 60 numbers $s_k$, $s_k + 6$ lie in $\{1, \ldots, 59\}$, so two are equal (Theorem 1.1); as each list is strictly increasing, $s_j = s_i + 6$ with $j > i$, and games $i + 1, \ldots, j$ have 6 goals.

(b) Goals $1, 1, 1, 1, 1, 7$ repeated five times: 30 games, 60 goals. A block containing a 7 has $\ge 7$ goals; otherwise it lies in a run of five 1's and has $\le 5$. ■

**Exercise 1.18 (Two 1's in north-east position)**

Let $n \ge 2$ and let $A$ be an $n \times n$ matrix with entries 0 and 1 only, at least $2n$ of which are 1. Prove that $A$ contains two entries equal to 1 such that one of them is strictly above and strictly to the right of the other.

**Solution.** Rows from the top, columns from the left: we need $(i', j')$, $(i, j)$ with $i' < i$, $j' > j$.

*Step 1 (boxes).* The $2n - 1$ anti-diagonals $D_s = \{(i, j) : i + j = s\}$, $s = 2, \ldots, 2n$, partition the positions.

*Step 2 (pigeon-hole).* $2n > 2n - 1$ ones, so two, $(i, j) \ne (i', j')$, lie in one $D_s$ (Theorem 1.1).

*Step 3 (conclusion).* Then $i \ne i'$, say $i' < i$, and $j' = s - i' > s - i = j$.

Sharpness: the $2n - 1$ ones of the last row and first column contain no such pair (for $(i, 1)$, $(n, j)$ the upper one is to the left). ■

**Exercise 1.35 (One number divides another)**

(Part (b) of the book's exercise.) Let $n \ge 1$. Prove that among any $n + 1$ distinct integers selected from $[2n]$ there are two such that one is a multiple of the other.

**Solution.**

*Step 1 (boxes).* Write uniquely $x = 2^a m$, $m$ odd (the odd part). For $x \in [2n]$, $m \in \{1, 3, \ldots, 2n-1\}$: $n$ values.

*Step 2 (pigeon-hole).* Two selected integers $x = 2^a m \neq y = 2^b m$ share $m$ (Theorem 1.1).

*Step 3 (conclusion).* $a \neq b$, say $a < b$; then $y = 2^{b-a} x$.

Sharpness: $n+1, \ldots, 2n$ contain no such pair ($2(n+1) > 2n$). ■

**Exercise 1.37 (Multiples of an irrational number)**

Let $r$ be an irrational real number. Prove that there is a positive integer $n$ such that the distance of $nr$ from the nearest integer is less than $10^{-10}$.

**Solution.** Write $N = 10^{10}$ and $\{x\} = x - \lfloor x \rfloor \in [0, 1)$ for the fractional part.

*Step 1 (boxes).* Split $[0, 1)$ into the $N$ half-open intervals $I_a = [\frac{a}{N}, \frac{a+1}{N})$, $a = 0, \ldots, N-1$.

*Step 2 (pigeon-hole).* Of the $N+1$ numbers $\{kr\}$, $0 \leq k \leq N$, two, $\{kr\}$ and $\{\ell r\}$ with $\ell < k$, share an $I_a$ (Theorem 1.1), so $|\{kr\} - \{\ell r\}| < \frac{1}{N}$.

*Step 3 (conclusion).* With $n = k - \ell \geq 1$, $M = \lfloor kr \rfloor - \lfloor \ell r \rfloor \in \mathbb{Z}$,

$$|nr - M| = |(kr - \lfloor kr \rfloor) - (\ell r - \lfloor \ell r \rfloor)| = |\{kr\} - \{\ell r\}| < 10^{-10}$$

(and $> 0$ by irrationality). ■